

Requirements TXT File Explanation

Dependency breakdown for `SkillUp_Hackathon_2026/requirements.txt`

What This File Is About: The `requirements.txt` file specifies all third-party Python packages required to run the project. Each entry enforces a minimum compatible version constraint (e.g., `>=`) to prevent dependency conflicts during installation via `pip install -r requirements.txt`.

PACKAGE ANALYSIS & DESCRIPTIONS

Package	Constraint	Purpose & Description
<code>streamlit</code>	<code>>= 1.32.0</code>	Powers the front-end web interface and interactive web dashboard entirely in Python.
<code>fpdf2</code>	<code>>= 2.7.9</code>	Programmatically builds, formats, and exports custom PDF documents.
<code>pandas</code>	<code>>= 2.0.0</code>	Handles tabular data manipulation, data cleaning, and CSV file reading/writing.
<code>segno</code>	<code>>= 1.6.1</code>	Generates high-quality standard QR codes and Micro QR codes without external dependencies.
<code>qrcode[pil]</code>	<code>>= 7.4.2</code>	Alternative QR code generator bundled with PIL hooks for custom raster image formatting.
<code>Pillow</code>	<code>>= 10.0.0</code>	Provides image processing utilities (opening, cropping, scaling, saving graphic formats).
<code>requests</code>	<code>>= 2.31.0</code>	Sends HTTP/1.1 requests to communicate with REST APIs and external web resources.
<code>yagmail</code>	<code>>= 0.15.293</code>	Simplifies sending automated Gmail notifications and email attachments.

Application Capability Summary

Based on these required libraries, the application performs the following workflow:

- Runs an interactive web UI using **Streamlit**.
- Processes data tables and datasets using **Pandas**.
- Generates QR codes using **Segno** and **QRCode (PIL)**.
- Builds downloadable or viewable PDF documents using **fpdf2** and **Pillow**.
- Queries web services or external endpoints using **Requests**.
- Dispatches automated email responses or generated PDF attachments using **Yagmail**.